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A Time Series Analysis-Based Stock Price Prediction Framework Using Artificial Intelligence

Harmanjeet Singh(✉) and Manisha Malhotra

University Institute of Computing, Chandigarh University, Sahibzada Ajit Singh Nagar (Mohali), Punjab, India
{harmanjeetsingh.uic,manisha.mca}@cumail.in

Abstract. Forecasting stock prices have recently emerged as an essential component of the economic realm. Stock price forecasting is regarded as a challenging endeavor due to the volatility and noise of stock market activity. In many stock price prediction scenarios, the Facebook Prophet, LightGBM and ARIMAX models have been demonstrated to be competitive versus other models. This research presents an architecture based on a time series model, such as Facebook Prophet, Light Gradient Boost Machine (GBM), and Autoregressive Integrated Moving Average with Explanatory Variable (ARIMAX) to accurately predict stock prices. Experiments with multiple potential outcomes are conducted to evaluate the suggested framework using the stock price data set. The model was trained on ADANI stock price data over the previous fourteen years using Facebook Prophet, LightGBM and ARIMAX and evaluated using the Root Mean Square Error metric (RMSE).

Keywords: Time series · Stock prediction · ARIMAX · Facebook Prophet · LightGBM

1 Introduction

In the past, stand-alone deep learning techniques were used to achieve the goal of predicting stock prices. But, employing the deep learning neural network in conjunction with state-of-the-art methodologies in the execution of iterative tasks delivers a more significant degree of success. In the present era of the internet, where ML (machine learning) and DL (deep learning) approaches are armed with the rigorous capability for study and forecasts regarding stock prices [1]. There are existing businesses and organizations operating within the financial sector that offer their services to commercial businesses, retail establishments, and financial institutions worldwide. This industry is comprised of institutions such as banks, financial businesses, insurance firms, and real estate groups, as mentioned in [2]. Traders can engage in transactions involving currencies, equities, and derivatives thanks to the virtual trading platforms made available by agents of financial markets. The actions of financial institutions profoundly influence the economic

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landscape. Because of the stock market, buying and selling publicly traded companies' shares can occur on or off specialized stock exchanges. Due to this market, even people who start with very few financial resources have the potential to amass a significant fortune. There is much interest in getting an accurate estimate of stock values in both the academic and corporate worlds, and for a good reason. Profits can be increased through stock price forecasting by academics and business specialists. A stock market is where stock transactions, trades, and distributions occur. Stocks must be moved about for any of these activities to occur. It has long been utilized by multinational corporations as one of their primary sources of finance [3]. Not only does the issuing of stocks bring in a considerable quantity of money, but it also fosters a greater concentration of funds, contributing to the consolidation of the organic structure of corporate capital. Predicting the price of a share of stock, although the stock market is notorious for its high levels of noise and its dynamic behaviors, is a highly challenging endeavor.

On the other hand, predicting stock prices requires extensive research and development time. Prices of stocks shift for several interrelated reasons, including geopolitical unrest, political maneuvering, shifts in economic conditions, shifts in market trends, advances in technology, and, perhaps most significantly, the attitudes of investors towards a particular stock. Time series or Deep learning is a new method that has recently come into existence [4], and is one of the numerous methods developed to anticipate the close price of a stock [5]. In this respect, the Long Short-Term Memory (LSTM) network, which intended to remember the temporal pattern for extended durations, is commonly employed for tasks involving time-series analysis [6, 4].

1.1 Contribution and Organization of Paper

In this work, a review of previous research on predicting stock prices using techniques based on artificial intelligence and hybrid models was done. The vast bulk of the methods that were proposed throughout the many publications was founded on the deep learning framework and hybrid outcomes for various stock market indexes. This study discusses the significant stock of a firm that uses AI-based time series approaches and the function of time series models with hyper-parameter tuning of features to get optimal outcomes.

The remaining parts of the article are organized as described below. In Sect. 2, detailed research methods are presented. This section includes the most current publication in stock price forecasting using AI-based time series models and methods. The architectures of the models that are being offered are discussed in Sect. 3. Section 4 presents the experiment research using data sets collected from the NSE website. Before moving on to the data pre-processing and model creation phase, this section widens the investigation of the technical indicators involved. In addition, this part provides a complete discussion of the model description, evaluation parameters, and outcomes after the section. The study comes to a close with a discussion of potential future options in Sect. 5.

2 Literature Review

In [7], Principal component analysis (PCA) was used to reduce the number of dimensions before the LSTM model was used to predict the stock's close price for the next trading day. This was done because technical indicators have a lot of dimensions. This was

done before applying the model to forecast the stock's close price for the following trading day. Compared to the LSTM standard model, it was revealed that the quantity of redundant data dropped, and the model achieved greater accuracy in its predictions. This was a significant finding. In a later step, an LSTM model without PCA and an LSTM model with PCA were compared to see how accurate and consistent the predictions were. This was done in order to determine whether or not the PCA-equipped LSTM model produced more accurate results.

In [2], ML and DL algorithms were applied to unstructured data gathered from social media and structured data in the form of international data and technical indicators to forecast the value of stocks. Most of this stock price projection was based on hybrid data acquired in areas where English is not the predominant language. Both LSTM and GA were utilized to determine the parameters for the stock market indexes. Based on evaluation criteria such as Mean Absolute Percentage Error (MAPE) and RMSE, these parameters were compared with those chosen by other models such as Back Propagation Neural Network Genetic Algorithm (BPNNGA), Least Squares Support Vector Regression Genetic Algorithm (LSSVRGA), Random Forest Genetic Algorithm (RFGA), and Extreme Gradient Boost Genetic Algorithm (XGBoostGA). By attaining MAPE values that varied from those of the other four models by a margin of less than five percent, the Long Short-Term Memory employed Genetic Algorithm (LSTMSGA) model produced exceptionally accurate prediction results.

In [3], a hybrid strategy that is both time and resource efficient, has been proposed to reduce the risk associated with investing during Black Swan occurrences. Experiments were carried out using the S\&P 500, the BSE Sensex, and the Nifty 50 as test subjects. It was claimed that the model achieved an accuracy of 86 percent during "black swan" events, which are unpredicted events that happen worldwide, such as a recession, a long disease, or war. The model was evaluated based on the accuracy parameter, and it was claimed that it achieved an accuracy of 80 percent for single step ahead prediction.

In [8], hybrid models of Bi-directional Long Short-Term Memory (Bi-LSTM) and Gated Recurrent Unit (GRU) were suggested, where first the individual performances of the Bi-LSTM model, the GRU model, and the standard neural model were investigated and assessed. Then a comparison was held between the individual models and the hybrid models. Model accuracy was determined using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Squared Log Error (MSLE), and Mean Absolute Percentage (MAPE) parameters.

In [9], it was recommended that the execution of the framework occur in two steps: selecting stocks for the portfolio and forecasting stock prices. Both of these phases would take place after the framework had been designed. Selecting stocks for a portfolio begins with constructing a mean-variance model using the stocks' historical data. It is followed by the determination of the sharp ratio. If the ratio is greater than one, the operation will continue to utilize Synthetic Minority Over-sampling (SMOTE) techniques. If this is not the case, a different group of stocks will be chosen.

In [10], LSTM and CNN-based hybrid model was presented, followed by an Attention block. Unfortunately, the model ran into the issue of over-fitting, which necessitated a change in the model's architecture in the form of an adjustment to the amount of neurons present in each layer and the addition of a drop-out layer after each of the layers.

It is essential to emphasise that accurate prediction relies heavily on effective attribute extraction, which is why this topic warrants special attention.

To efficiently learn the dynamics for share price from a data-set, [11] offers a hybrid DL framework combine with CNNs model and LSTM model. The first step taken into consideration due to a time series data set containing stock prices was the extraction of various time scales from the data set using the CNN layer. The original prices were blended at a later date using time scale extraction. It was also thought about how to use multiple LSTMs to understand temporal dependencies for different time-scale features. While attempting to anticipate the close price of a stock, all of the features that have been gathered by LSTMs are integrated using linked neuron networks. Still, the above strategy doesn't fully understand how the interdependence of stock prices that are very long doesn't work.

In [12], it was possible to do a comparative examination of twenty-five distinct ensembles' worth of regressors and classifiers. The study's results showed that stacking and mixing ensembles worked better than bagging and boosting. While evaluating the models, the RMSE parameter was considered, and it was reported that stacking obtained 0.0001–0.001 and blending achieved 0.002–0.01, both of which were deemed to be superior to other ensemble strategies.

In [13], the authors described hybrid deep learning models capable of forecasting and/or analyzing time series data-sets. These models were made using multi-scale features, and the authors used many pipelines or networks to pull out the multi-scale features. This resulted in a large and complex model that was not considered favorable for training or testing and was also irrelevant to insight patterns.

A new hybrid approach introduced in [14], LSTM auto-encoder and stacked LSTM network comprise the first and second layers of the LSTM model's fundamental structure, respectively. An auto-encoder method was observed to be quite beneficial for noise reduction in inputted data, which results in negligible mistakes in the case of a basic long short term memory model and a standard multi-layer perceptron. This finding was made after it was discovered that an auto-encoder method had shown quite beneficial results. The conclusion was reached when more errors were produced when the parameter had a high value.

Ensemble model of LSTM networks for Intraday trading forecasting was proposed in [15], in which a series of LSTM network models are used individually before being joined in series to evaluate overall performance. The network inputs are taken from a broad variety of technical analysis indicators. The results showed that using Ensemble LSTM significantly improved over typical ensemble weighting procedures. For example, a model with an equally weighted technique, or one with an ideal single model, may be evaluated based on its power to forecast the price of 22 stocks out of 44 US big-cap businesses. Additionally, the outcomes of the ensemble LSTM model are superior to those of standalone models like the lasso and ridge logistic regression. Although surpassing several different benchmarks, the recommended solution has a significantly higher overall cost in terms of computation. The training duration of the ensemble model for each stock took roughly one hundred and fifty minutes, whereas the training period of lasso for the same job just took a few seconds.

Multi-filter neural networks such as recurrent and convolutional networks were proposed in [16], to extract features from time series data. These hybrid neural networks outperformed single-structure neural networks when put in comparison with state of art parameters. Although it is an inevitable fact that for short term duration learning, CNNs models are excellent whereas Recurrent Neural Networks (RNN) are considered to be excellent in long term duration. However, in [17] networks were observed to struggle in comprehend the extremely long stock price data-sets.

Despite this, the time series that we have provided considers all of the problems that have been brought up to correctly predict stock prices. ARIMAX, Facebook Prophet, and LightGBM are the three components that make it up.

3 Architecture of Proposed Models

In this research, time series models were deployed on stock named as Adani Port to forecast stock price. Initially, the dataset was preprocessed. Further, three windows were designed, one for every three days, one for every seven days, and one for every thirty days, with features corresponding to "high," "low," "volume," "turnover," and "trades." For each time window, average of attributes was calculated. Then, for each time window, standard deviation was calculated. After that, the dataset was split into two subsets and the mean value of each feature was put in the empty cells of each subset. The data collected before 2019 served as our training data set, while the data collected after 2019 served as our test data set. Because stock prices are highly impacted by noise and disturbance from the outside, in order to stabilize this noise, we include a list of exogenous features. These features include high mean, low mean, high standard deviation, low standard deviation, volume mean, volume standard deviation, turnover mean, and turnover standard deviation in durations of a day, day of the week, a week, and a month.

After this, the proposed model is trained with AutoARIMA, FBProbhet and LightGBM. The following steps were carried out in suggested framework's stock prediction process:

a) **Data collection:** Stock data is gathered from yahoo finance from the period of Nov. 2007 to April. 2021 and supplied into the pre-processing step during this stage.
b) **Data Pre-processing:** In this phase, involves transforming raw data into something that can be analysed.
c) **Data Modeling:** a stock prediction model is built and the converted data are input into it. To properly anticipate the closing price of a stock, data analysis is carried out in this stage.
d) **Performance Evaluation:** It is here that the model's output is compared against actual output in order to determine whether or not the model is accurate. Model improvement or data preparation enhancement is needed if this model does not meet its stated goals. Otherwise, inform the user of the findings.
e) **Result:** Once the model has been shown, the user may see the findings and predictions.

4 Experiment Study

Figure 1 shows the five critical features from data-set that can be use in prediction of stock movement. These features are stock's open price, close price, high price, low price and last close price.

4.1 Dataset

This analysis makes use of the historical data on the stock that Yahoo Finance generated. Figure 1 illustrates an example of this data collection, which has a variety of attributes such as open (the stock's current open price), close (its current close price), high (the stock's highest price in a day), low (the stock's lowest price in day), close (the stock's closing price in a day), and last (last close price of a stock).

Fig. 1. Original data-set of Adani port's share.

The direction of the stock movement is reflected in the adjusted characteristics to provide an accurate depiction of a company's value after considering its operational operations. This data set provides information on the price movement of Adani stock's throughout the given period. Figure 1 depicts Adani's share price close index.

One set of data is used for training, while the other is utilized for testing, for two sets of data. The train data set accounts for 70 percent of the source data set, whereas the test data set comprises 30 percent. Data ranging from November 27, 2007 (which is a Tuesday) through April 19, 2017 (which is a Wednesday) was used to train the models. The models are then assessed using data ranging from Thursday, April 20, 2017, until Thursday, April 30, 2021.

4.2 Technical Indicators

For the stock price of Adani Port, an average directional movement was determined where ADX $< = 25$ is considered not in trends, ADX $= > 25$ to ADX $< = 50$ is considered in trends and ADX $= > 50$ is considered in high trends. It performs well in determining the direction of a trend, which is useful for market analysts and investors (Fig. 2).

Fig. 2. Technical indicator as average directional movement of Adani port's share price

The other type of technical indicator that we employed in our research was called the relative strength index (RSI). It illustrates how quickly and with how much momentum stock prices have been moving in recent times (as seen in Fig. 3). As a consequence, the stock price may be judged to be overbought or oversold depending on certain criteria. Generally, a stock is considered overbought when its RSI index is higher than 70, while

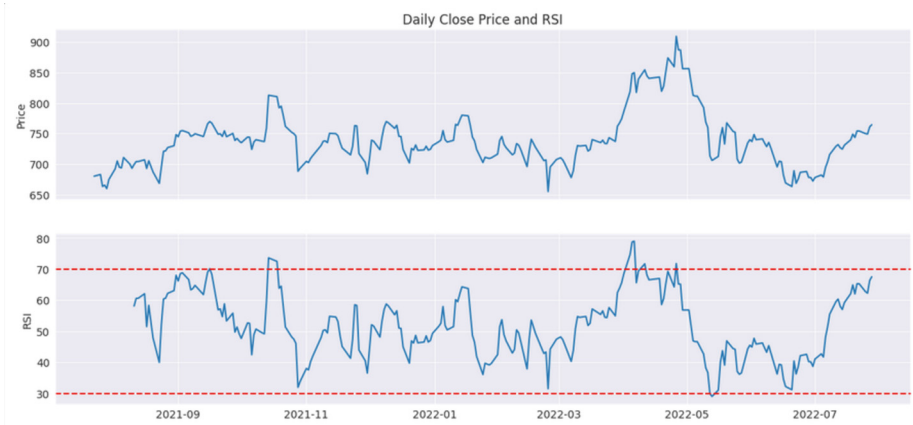

Fig. 3. Technical indicator as RSI index of Adani port's share price

a stock is considered to be oversold when its RSI index is lower than 30. Further information from the RSI indicator suggests that stock prices may be getting ready to make a trend reversal or corrective retreat. This index, in general, assists the investor in making decisions on the optimal moment to purchase or sell shares. Figure 3 illustrates Adani port's daily close price as well as the RSI index over the specified period.

4.3 Data Pre-processing and Model Design

In order to create the model of the time series, we followed the process that is mentioned below:

Following the separation of the data set into a train set of data before the year 2019 and a test set of data after the year 2019, based on the close price column, we processed the training and test data in such a way that every next day's close price is dependent on the different windows size such as 3-days, 7-days, and 30-days the previous values of stock's open price, close price, high price, low price, and last close price, which are then marked as features. This was accomplished Following the completion of the process involving the preparation of the data, we computed the mean and standard deviation of each characteristic in order to arrive at a point where we could validate the presence of stationary data. In order to create our models, we relied on the library resources provided by the Prophet, PMDARIMA, and LightGBM. In order to develop the AUTO ARIMA model, we used the VWAP (volume average weighted price) of the train dataset in conjunction with some exogenous characteristics. The model is constructed with the loss parameter set to "root mean squared error," and it also has the "mean absolute error" setting.

In order to develop the Facebook Prophet model and LightGBM, date and close price columns from training data were taken into consideration with VWAP columns.

4.4 Result and Discussion

In order to successfully implement all of the time series models, the frameworks TensorFlow 2.8.0 and Keras 2.8.0 were utilized. The researchers used a 64-bit Windows 10, an Intel i5-8265U central processing unit (CPU), 16 gigabytes of random-access memory (RAM), and 64-bit Windows to train the models. In the instance of Auto ARIMA, it took around 210 ms, generating an RMSE of training data of 27.12, followed by an MAE of training data of 10.61. In the instance of LightGBM, it took around 213 ms, giving an RMSE of 89.98 for the training data, followed by an MAE of 44.76 for the training data. In Facebook prophet, it took around 202 ms, generating an RMSE of training data of 29.72, and MAE of training data of 16.26.

The trained model is given test data so that the "Loss," "Mean Squared Error," and "Mean Absolute Error" parameters can be used to measure how well it can predict outcomes. This is achieved by utilizing the mean squared error library, which is imported from the sklearn package. The model generated the results that are displayed in Table 1.

The graph of the same is shown in Fig. 4 where the ARIMAX forecast line gives better prediction results than other models.

Table 1. Evaluation matrices of prediction result on training data in time series models.

Stock Name	Model's evaluation of training records		
	Model	RMSE	MAE
Adani Port	ARIMAX	27.12	10.61
	LightGBM	89.98	44.76
	FBProphet	29.72	16.26

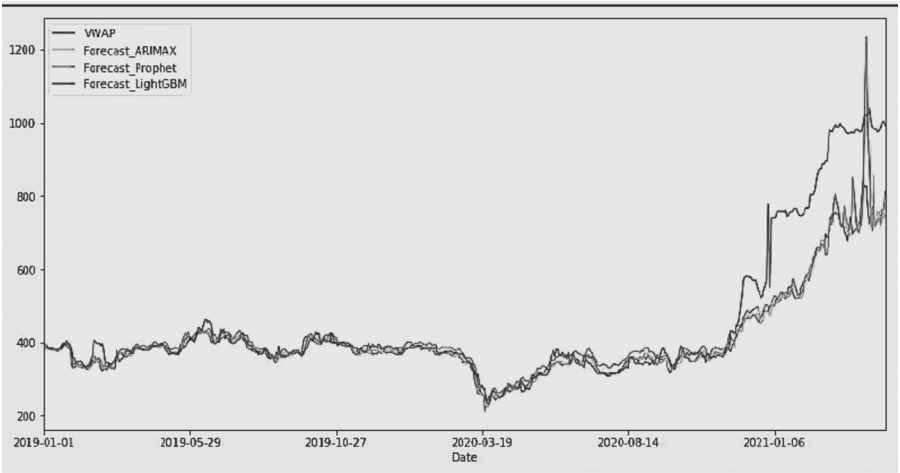

Fig. 4. Predicted price of Adani port's share.

5 Conclusion

Computerized financial market price forecasting is attracting investors. Consequently, ARIMAX, FBProphet, and LightGBM time series models may predict stock prices. In this study, we invented a time series-based stock market forecasting approach. The suggested model calculated ADANI Ports' pricing with the lowest RMSE and MAE. Our goal is to use classical, ensemble, and time series methods together to make more reliable future work models.

In future research, the suggested technique may be applied to stock market data from different nations, and comparisons can be made with XGBoost, SVM, and ANN. The SVM and ANN models, on the other hand, can be built using several different things, such as investor mood, interest rates, the country's political situation, news, and currency rates. In addition, we will assess the accuracy of our proposed model's predictions for various nonlinear and non-stationary time series, including exchange rates, crude oil prices, wind speed, temperature, precipitation, earthquakes, and tourist arrivals.

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